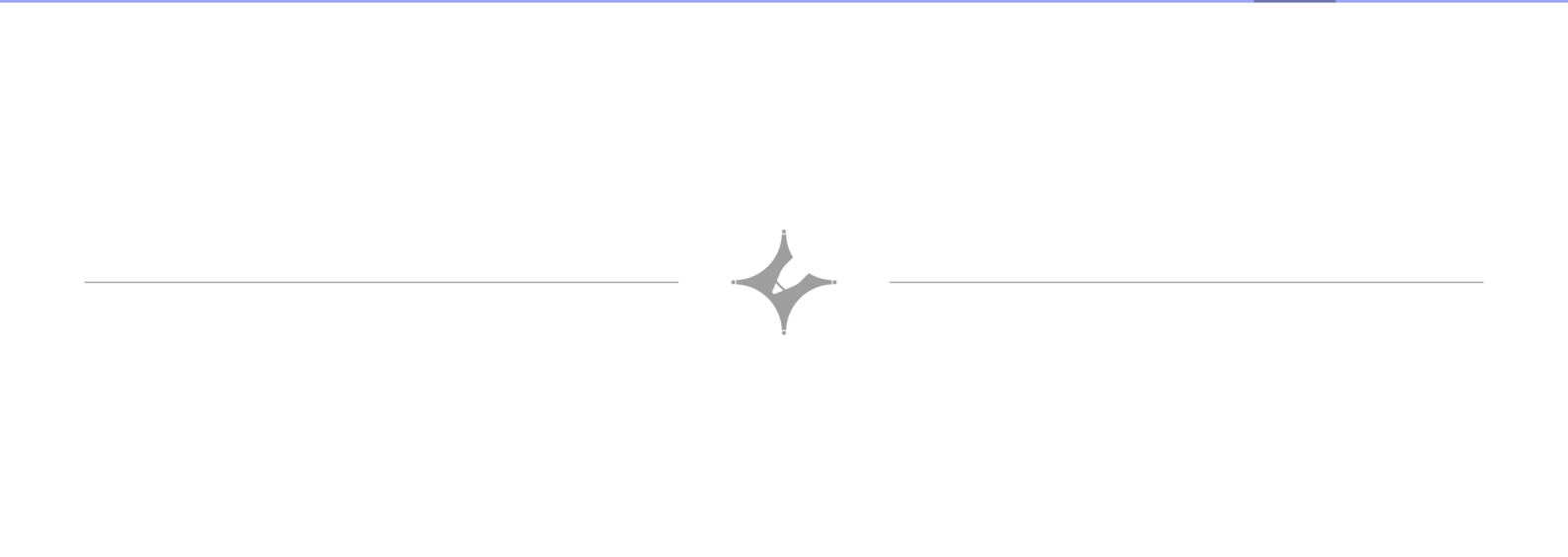
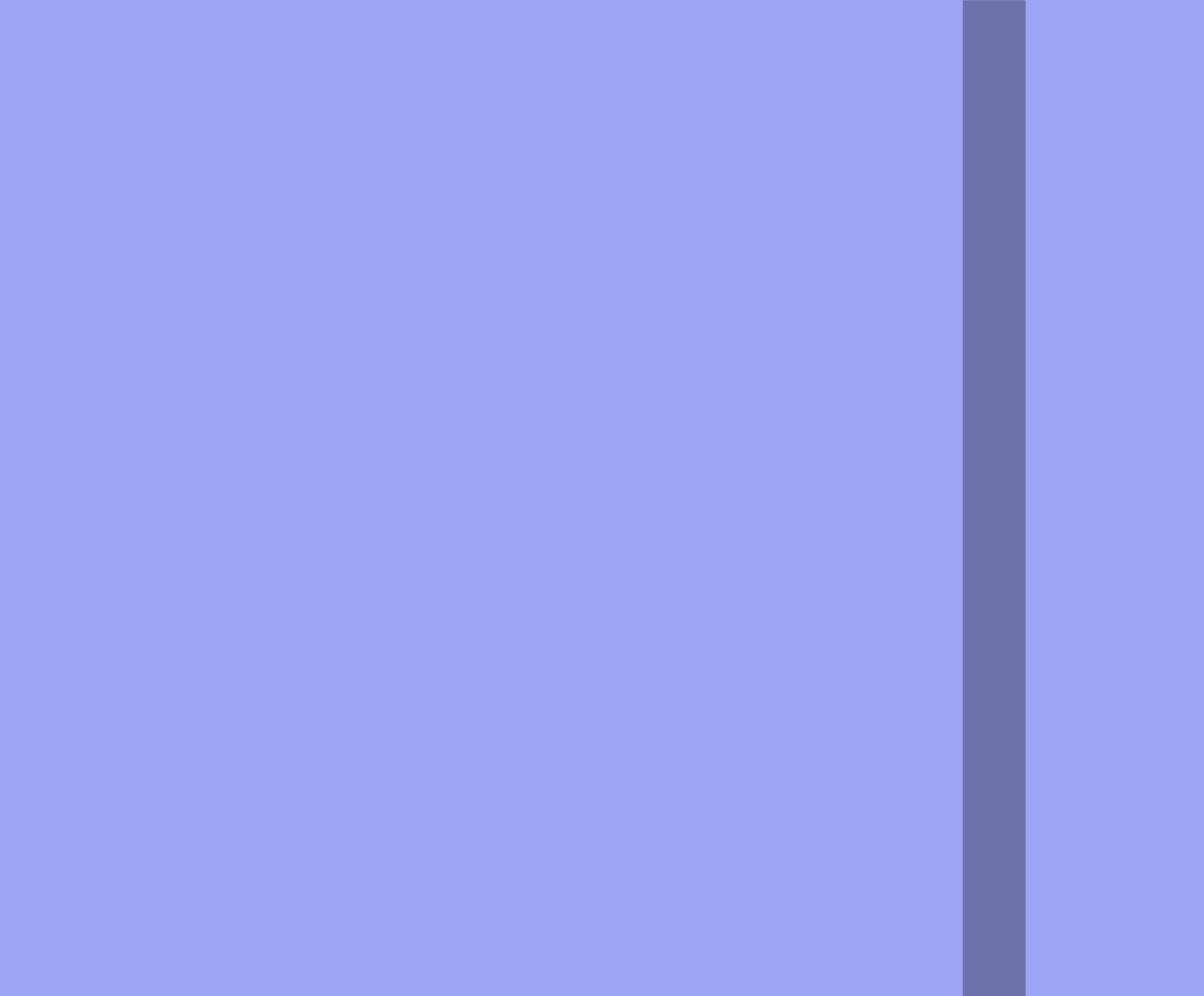




Pump Theorem.

Let L Regular. $\exists p \geq 1$, s.t. $\forall w \in L$ with $|w| \geq p$

可被分为3部分 $w = xyz$, 满足

(1). $\forall i \geq 0, xy^i z \in L$

(2). $|y| > 0$

(3). $|xy| \leq p$.

pumping length

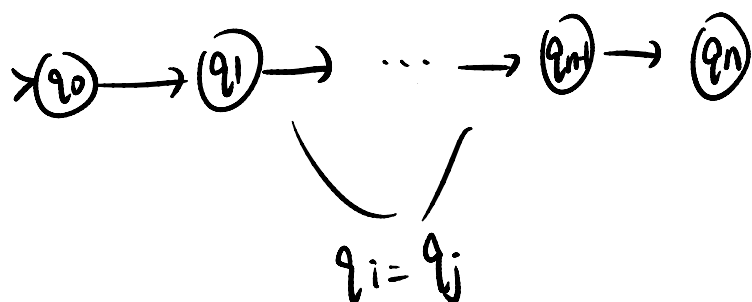
例: ab^*a . $p=2 : aa \quad \times$
 $p=3 : aba \quad \checkmark$

下证 pump theorem.

$\exists$ DFA M decide L .

令 $p = \# \text{ states of } M$

若 $w \in L$ with $|w| \geq p$.



$n \geq p$.

$\therefore$ 必有 $0 \leq i < j \leq p$, s.t. $q_i = q_j$. (抽屉原理)

令前、中、后为 $x y z$, 显然 $x y^i z \in L, \forall i$ ①

$$|y| = j - i > 0 \quad (2).$$

$$|xy| = j \leq p \quad (3).$$

#.

例: $0^n 1^n$ 非正则 (记为 A)

设 A 正则

$\therefore$ let p 为 pump length.

let $w = 0^p 1^p$, $w = xyz$ 满足3个条件.

由(2)、(3), $y = 0^k$, $x y^p z = 0^{p-k} 1^p \notin A$

矛盾 $\therefore 0^n 1^n$ 非正则.

#.

